

取 7 个节点的函数值，分别利用复化的梯形公式和复化 Simpson 公式计算积分

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{x} dx \text{ 的近似值 } \left(\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 \right).$$

1. 解:

$$0 \quad \frac{\pi}{12} \quad \frac{\pi}{6} \quad \frac{3\pi}{12} \quad \frac{\pi}{3} \quad \frac{5\pi}{12} \quad \frac{\pi}{2}$$

(1) 复化梯形公式 ($n=6$)

$$\begin{aligned} T &= \frac{b-a}{2n} \left[f(a) + 2 \sum_{k=1}^{n-1} f(x_k) + f(b) \right] \\ &= \frac{\pi}{24} \times \left[f(0) + 2 \times \left(f\left(\frac{\pi}{12}\right) + f\left(\frac{\pi}{6}\right) + f\left(\frac{3\pi}{12}\right) + f\left(\frac{\pi}{3}\right) + f\left(\frac{5\pi}{12}\right) \right) + f\left(\frac{\pi}{2}\right) \right] \\ &= \frac{\pi}{24} \times \left[1 + 2 \times \left(\frac{12}{\pi} \sin \frac{\pi}{12} + \frac{6}{\pi} \sin \frac{\pi}{6} + \frac{4}{\pi} \sin \frac{\pi}{4} + \frac{3}{\pi} \sin \frac{\pi}{3} + \frac{12}{5\pi} \sin \frac{5\pi}{12} \right) + \frac{2}{\pi} \right] \\ &= 1.368 \end{aligned}$$

(2) 复化 Simpson 公式 ($n=3$)

$$\begin{aligned} S &= \frac{b-a}{6n} \left[f(a) + 2 \sum_{k=1}^{n-1} f(x_k) + 4 \sum_{k=1}^{n-1} f\left(x_{k+\frac{1}{2}}\right) + f(b) \right] \\ &= \frac{\pi}{36} \times \left[1 + 2 \times \left(f\left(\frac{\pi}{6}\right) + f\left(\frac{\pi}{3}\right) \right) + 4 \times \left(f\left(\frac{\pi}{12}\right) + f\left(\frac{3\pi}{12}\right) + f\left(\frac{5\pi}{12}\right) \right) + f\left(\frac{\pi}{2}\right) \right] \\ &= \frac{\pi}{36} \times \left[1 + 2 \times \left(\frac{6}{\pi} \sin \frac{\pi}{6} + \frac{3}{\pi} \sin \frac{\pi}{3} \right) + 4 \times \left(\frac{12}{\pi} \sin \frac{\pi}{12} + \frac{4}{\pi} \sin \frac{\pi}{4} + \frac{12}{5\pi} \sin \frac{5\pi}{12} \right) + \frac{2}{\pi} \right] \\ &= 1.371 \end{aligned}$$

确定求积公式

$$\int_{-1}^1 f(x) dx \approx Af(-1) + Bf(x_1)$$

中的待定系数，使其代数精度尽可能高，并指出其代数精度。

2. 解。

式中待定系数有3个，故需3个方程。

分别取 $f(x) = 1, x, x^2$ 得：

$$\begin{cases} A+B=2 \\ -A+Bx_1=0 \\ A+Bx_1^2=\frac{2}{3} \end{cases} \Rightarrow \begin{cases} A=\frac{1}{2} \\ B=\frac{3}{2} \\ x_1=\frac{1}{3} \end{cases} \quad \leftarrow \text{用代数精度定义的}$$

$$\text{故 } \int_{-1}^1 f(x) dx \approx \frac{1}{2} [f(-1) + 3f(\frac{1}{3})]$$

$$\text{当 } f(x) = x^3 \text{ 时, } \frac{1}{2} f(-1) + \frac{3}{2} f(\frac{1}{3}) = -\frac{1}{2} + \frac{3}{2} \times \frac{1}{27} = -\frac{4}{9} \neq 0 = \int_{-1}^1 x^3 dx$$

$\therefore A = \frac{1}{2}, B = \frac{3}{2}, x_1 = \frac{1}{3}$ 时，代数精度最高为2次

$w_0(x), w_1(x), \dots, w_n(x), \dots$ 是 $[a, b]$ 上以 $\rho(x)$ 为权函数的标准正交多项式系,

$x_i (i = 0, 1, \dots, n)$ 是 $w_{n+1}(x)$ 的零点, ω

$$\int_a^b \rho(x) f(x) dx \approx \sum_{k=0}^n A_k f(x_k) \quad \omega$$

是以 $x_i (i = 0, 1, \dots, n)$ 为节点的 Gauss 型求积公式, 证明: 当 $0 \leq i < j \leq n$ 时,

$$\sum_{k=0}^n A_k \omega_i(x_k) \omega_j(x_k) = 0. \quad \omega$$

3. 证明:

$\therefore$ Gauss 型求积公式具有 $2n+1$ 次代数精度

$\therefore$ 对于 $t \leq 2n+1$, 若 $f(x) \in P_t$, $\int_a^b \rho(x) f(x) dx = \sum_{k=0}^n A_k f(x_k)$

取 $f(x) = w_i(x) w_j(x)$, $0 \leq i < j \leq n$, 则 $i+j \leq 2n < t \leq 2n+1$

$\therefore f(x) \in P_t$, $t \leq 2n+1$, 故

$$\int_a^b \rho(x) f(x) dx = \int_a^b \rho(x) w_i(x) w_j(x) dx = \sum_{k=0}^n A_k f(x_k) = \sum_{k=0}^n A_k w_i(x_k) w_j(x_k)$$

$\therefore w_i(x)$ 与 $w_j(x)$ 是 $[a, b]$ 上以 $\rho(x)$ 为权函数的正交多项式

$$\therefore \int_a^b \rho(x) w_i(x) w_j(x) dx = \sum_{k=0}^n A_k w_i(x_k) w_j(x_k) = 0$$

利用 Gauss-Legendre 公式构造 Gauss 型求积公式。

$$\int_0^1 f(x) dx \approx A_0 f(x_0) + A_1 f(x_1).$$

4. 解:

$$\text{Gauss-Legendre 公式为 } \int_{-1}^1 f(x) dx \approx f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$$

$$\text{令 } x = \frac{1}{2} + \frac{1}{2}t, \text{ 则 } \int_0^1 f(x) dx = \frac{1}{2} \int_{-1}^1 f\left(\frac{1}{2} + \frac{1}{2}t\right) dt$$

$$\text{可得 } \int_{-1}^1 f\left(\frac{1}{2} + \frac{1}{2}t\right) dt \approx f\left(\frac{1}{2} - \frac{1}{2\sqrt{3}}\right) + f\left(\frac{1}{2} + \frac{1}{2\sqrt{3}}\right)$$

$$\text{故 } \int_0^1 f(x) dx \approx \frac{1}{2} \left[f\left(\frac{1}{2} - \frac{1}{2\sqrt{3}}\right) + f\left(\frac{1}{2} + \frac{1}{2\sqrt{3}}\right) \right]$$

利用三点 Gauss-Chebyshev 求积公式计算定积分： $\int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx$ 的近似值。

5. 解：

$$n=2, \quad k=0, 1, 2; \quad f(x)=x^2$$

$$k=0 \text{ 时}, \quad x_0 = \cos\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} \quad A_0 = \frac{\pi}{3}$$

$$k=1 \text{ 时}, \quad x_1 = \cos\left(\frac{\pi}{2}\right) = 0 \quad A_1 = \frac{\pi}{3}$$

$$k=2 \text{ 时}, \quad x_2 = \cos\left(\frac{5\pi}{6}\right) = -\frac{\sqrt{3}}{2} \quad A_2 = \frac{\pi}{3}$$

$$\therefore \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx \approx \frac{\pi}{3} \times \frac{3}{4} + \frac{\pi}{3} \times 0 + \frac{\pi}{3} \times \frac{3}{4} = \frac{\pi}{2}$$

用三点公式求 $f(x) = \frac{1}{(1+x)^2}$ 在 $x=1.0, 1.1, 1.2$ 处的导数值. 已知

$$f(1.0) = 0.250000, f(1.1) = 0.226757, f(1.2) = 0.206612.$$

解:

$$f(1.0) = 0.250000 \quad f(1.1) = 0.226757 \quad f(1.2) = 0.206612$$

$$f'(1.0) = \frac{1}{2 \times 0.1} \times (1 - 3 \times 0.25 + 4 \times 0.226757 - 0.206612) = -0.24792$$

$$f'(1.1) = \frac{1}{2 \times 0.1} \times (0.206612 - 0.25) = -0.21694$$

$$f'(1.2) = \frac{1}{2 \times 0.1} \times (0.25 - 4 \times 0.226757 + 3 \times 0.206612) = -0.18596$$

确定下列插值型求积公式中的待定系数，并求其代数精度

$$\int_{-1}^1 f(x) dx \approx A_0 f(-1) + A_1 f(0) + A_2 f(1)。$$

7. 解=

取 $f(x) = 1, x, x^2$ ，则

$$\begin{cases} A_0 + A_1 + A_2 = \int_{-1}^1 1 dx = 2 \\ -A_0 + 0 + A_2 = \int_{-1}^1 x dx = 0 \\ A_0 + 0 + A_2 = \int_{-1}^1 x^2 dx = \frac{2}{3} \end{cases} \quad \text{得} \quad \begin{cases} A_0 = \frac{1}{3} \\ A_1 = \frac{4}{3} \\ A_2 = \frac{1}{3} \end{cases}$$

$$\therefore \int_{-1}^1 f(x) dx \approx \frac{1}{3} f(-1) + \frac{4}{3} f(0) + \frac{1}{3} f(1)$$

$$\text{当 } f(x) = x^3, \frac{1}{3} f(-1) + \frac{4}{3} f(0) + \frac{1}{3} f(1) = -\frac{1}{3} + \frac{1}{3} = 0 = \int_{-1}^1 x^3 dx$$

$$\text{当 } f(x) = x^4, \frac{1}{3} f(-1) + \frac{4}{3} f(0) + \frac{1}{3} f(1) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \neq \frac{2}{5} = \int_{-1}^1 x^4 dx$$

$\therefore A_0 = \frac{1}{3}, A_1 = \frac{4}{3}, A_2 = \frac{1}{3}$ ，代数精度为3次